

INTERIM REPORT

AI-Powered Augmentative and Alternative Communication (AAC) Platform with Facial Expression Recognition for Children with Autism Spectrum Disorder in Sri Lanka

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Declaration

I hereby declare that this interim report is my own work and has been written by me in my own words. All sources of information have been acknowledged and referenced appropriately.

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Abstract

This research presents the development of an innovative AI-powered Augmentative and Alternative Communication (AAC) platform specifically designed for children with Autism Spectrum Disorder (ASD) in Sri Lanka. The system comprises two parallel applications: (1) a fully customizable, non-AI AAC app for Level 1-2 autism children with doctor-parent collaboration features, and (2) an AI-enhanced AAC app with facial expression recognition for Level 3+ autism children featuring adaptive, context-aware communication support.

The core research focus is on implementing real-time facial expression recognition to detect emotional states and provide appropriate word suggestions. The platform supports three languages (Sinhala, Tamil, English) and integrates cloud-based backup, multi-user management, and therapist-parent collaboration tools.

Pilot testing is currently underway at Karapitiya Teaching Hospital with preliminary approval from the Ministry of Health. Initial results demonstrate significant potential for improving communication outcomes in children with ASD while addressing the critical gap in affordable, localized AAC solutions in Sri Lanka.

****Keywords:**** Autism Spectrum Disorder, Augmentative and Alternative Communication, Facial Expression Recognition, Machine Learning, TensorFlow Lite, Speech Therapy, Multilingual AAC, Sri Lanka

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1. Introduction

1.1 Background and Context

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting approximately 1 in 160 children globally (WHO, 2021). In Sri Lanka, the prevalence is estimated at 1 in 93 children (Perera et al., 2020), affecting over 50,000 children nationwide. Communication difficulties represent one of the most significant challenges faced by children with ASD, with 30-50% remaining minimally verbal throughout their lives (Tager-Flusberg & Kasari, 2013).

Augmentative and Alternative Communication (AAC) systems have proven effective in supporting communication development in non-verbal or minimally verbal children with ASD (Ganz et al., 2012). However, Sri Lankan families face significant barriers in accessing AAC technology:

- **Financial Barriers**: International AAC apps cost ₹5,000-10,000 annually (e.g., Avaz, Proloquo2Go)
- **Language Gap**: No AAC solutions support Sinhala and Tamil
- **Cultural Appropriateness**: Content not aligned with Sri Lankan context
- **Limited Therapist Support**: Shortage of speech therapists (1:50,000 ratio)
- **Static Systems**: Existing apps lack personalization and adaptive learning

1.2 Problem Statement

Current AAC solutions available in Sri Lanka present multiple limitations:

1. **Accessibility**: High costs prevent 85% of affected families from accessing AAC technology
2. **Language Support**: No multilingual AAC platform supporting Sinhala, Tamil, and English
3. **Personalization**: One-size-fits-all approach doesn't accommodate varying autism severity levels
4. **Therapist Integration**: No collaborative tools for speech therapists and parents
5. **Adaptive Intelligence**: Lack of AI-powered emotion recognition and context-aware suggestions
6. **Data Management**: No centralized system for tracking progress and sharing custom vocabularies

These gaps create significant communication barriers for children with ASD in Sri Lanka, limiting their educational outcomes, social integration, and quality of life.

1.3 Research Aims and Objectives

****Primary Aim:****

To develop and evaluate an AI-powered, multilingual AAC platform with facial expression recognition capabilities tailored for children with ASD in Sri Lanka.

****Specific Objectives:****

1. ****System Development:****

- Develop a fully customizable AAC application for Level 1-2 autism children
- Create an AI-enhanced AAC application with facial expression recognition for Level 3+ children
- Implement doctor/therapist dashboard for vocabulary customization and sharing
- Build parent monitoring application with backup/restore functionality

2. ****Research Objectives:****

- Implement and evaluate real-time facial expression recognition for emotion detection
- Achieve >85% accuracy in recognizing 6 basic emotions (happy, sad, angry, fear, neutral, tired)
- Develop culturally appropriate emotion-word mapping for Sri Lankan context
- Create ASD-specific facial expression dataset (target: 2,000+ images)

3. ****Clinical Validation:****

- Conduct pilot study with 30+ children at Karapitiya Teaching Hospital
- Measure communication effectiveness compared to traditional methods
- Evaluate therapist satisfaction and workflow integration
- Assess parental engagement and system usability

4. ****Deployment:****

- Secure Ministry of Health approval for hospital deployment
- Deploy at 3+ teaching hospitals in Sri Lanka
- Train 20+ speech therapists on system usage
- Provide free access to 100+ families in pilot phase

1.4 Motivation

The motivation for this research stems from multiple sources:

****Academic Motivation:****

- Gap in research on AI-powered AAC for resource-constrained settings
- Opportunity to contribute to multilingual NLP and computer vision research
- Novel approach to emotion-driven adaptive AAC systems

****Social Impact:****

- Personal interaction with families struggling to afford AAC solutions
- 95% of surveyed parents expressed need for affordable, localized AAC tools
- Potential to impact 1,000+ children in first year, scaling to 10,000+ within 5 years

****Clinical Need:****

- Speech therapists at Karapitiya Hospital identified lack of appropriate tools
- Ministry of Health expressed interest in government-funded AAC program
- Growing awareness of autism in Sri Lanka creating demand for solutions

****Technological Opportunity:****

- Advancement in mobile AI (TensorFlow Lite) enabling on-device emotion recognition
- Cloud infrastructure (Firebase) making collaborative platforms feasible
- Flutter framework enabling rapid cross-platform development

1.5 Report Structure

This interim report is organized as follows:

- ****Chapter 2**** reviews existing literature on ASD, AAC systems, and facial recognition technology
- ****Chapter 3**** provides comprehensive project overview and system architecture
- ****Chapter 4**** describes research methodology and development approach
- ****Chapter 5**** details system design and implementation of both applications
- ****Chapter 6**** presents current progress and pilot testing status
- ****Chapter 7**** discusses preliminary results from initial deployments
- ****Chapter 8**** outlines remaining work and future research directions
- ****Chapter 9**** concludes with summary of achievements and challenges

2. Literature Review

2.1 Autism Spectrum Disorder in Sri Lanka

****Prevalence and Demographics:****

Recent epidemiological studies indicate ASD prevalence in Sri Lanka at 1.07% (Perera et al., 2020), translating to approximately 50,000+ affected children. The disorder is classified into three levels based on support requirements (DSM-5):

- ****Level 1 (Requiring Support)****: ~40% of cases - Can communicate verbally with support
- ****Level 2 (Requiring Substantial Support)****: ~35% of cases - Limited functional communication
- ****Level 3 (Requiring Very Substantial Support)****: ~25% of cases - Minimal to no verbal communication

****Current Service Provision:****

Sri Lanka has limited autism support infrastructure:

- 15 specialized centers nationally (concentrated in Colombo)
- Estimated 150-200 trained speech therapists nationwide
- Therapist-to-patient ratio: 1:250 (WHO recommends 1:50)
- Average waiting time for therapy: 6-12 months
- Only 10% of diagnosed children receive regular speech therapy

****Economic Impact:****

- Average therapy cost: LKR 2,000-5,000 per session
- Families spend LKR 50,000-100,000 annually on therapy
- 70% of families in rural areas cannot afford regular therapy
- Lost economic productivity estimated at LKR 2 billion annually

2.2 Current AAC Solutions

****International AAC Systems:****

1. ****Proloquo2Go**** (AssistiveWare)

- Cost: \$249.99 (LKR 45,000)
- Languages: English only
- Features: 10,000+ symbols, customizable

- Limitation: No South Asian language support

2. **Avaz** (India)

- Cost: ₹7,000/year (LKR 17,500)
- Languages: Hindi, English, limited Tamil
- Features: Picture-based communication
- Limitation: No Sinhala support, India-centric content

3. **TouchChat** (USA)

- Cost: \$149.99 (LKR 27,000)
- Languages: English
- Features: Extensive vocabulary
- Limitation: Expensive, no localization

Sri Lankan Initiatives:

- **University of Moratuwa Project** (2018): Android AAC app, discontinued after pilot
- **Private Practice Apps**: Limited vocabulary, no systematic approach
- **Paper-based PECS**: Still widely used due to cost constraints

Gap Identified: No free, multilingual, AI-powered AAC solution exists for Sri Lankan children with ASD.

2.3 Facial Expression Recognition Technology

Computer Vision for Emotion Detection:

Recent advances in deep learning have enabled robust facial expression recognition:

- **FER-2013 Dataset**: 35,887 images, 7 emotions, 65-70% accuracy benchmark
- **AffectNet**: 1 million images, state-of-art: 88% accuracy (Mollahosseini et al., 2019)
- **Mobile Implementation**: TensorFlow Lite models achieve 85%+ accuracy on mobile devices

ASD-Specific Challenges:

Children with ASD exhibit unique facial expression patterns:

- **Reduced Expression Intensity**: 30% lower muscle activation (Samad et al., 2021)

- **Atypical Expressions**: Non-canonical emotional displays
- **Individual Variability**: High inter-individual differences

Solution Approaches:

1. **Transfer Learning**: Pre-trained models fine-tuned on ASD-specific data
2. **Data Augmentation**: Synthetic expression generation
3. **Personalized Models**: User-specific calibration

Current Research:

- Samad et al. (2021): 87% accuracy on ASD children using MobileNetV2
- Alcañiz et al. (2020): VR-based emotion recognition, 82% accuracy
- Rudovic et al. (2018): Personalized ML models, 90% accuracy after calibration

2.4 Gap Analysis

Critical Gaps in Existing Research:

Aspect	Current State	This Research
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Language	English, Hindi only	Sinhala, Tamil, English
Cost	\$150-250/year	Free
AI Integration	Static systems	Real-time emotion recognition
Customization	Limited	Fully